

Documentary Inertia vs. Evolutionary Fitness:

What Talkie-1930

Reveals About the Validity Limits of Memetic Persistence Indices

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ABSTRACT

The Extended Phenotype Theory of law (EPT) treats legal and institutional norms as memes subject to evolutionary selection: what persists in the normative corpus is what survived the selection pressures of its environment. The Constitutional Lock-in Index (CLI) and the Transmission Capacity Index (TCI), developed in prior work within the EPT programme, operationalize memetic fitness as persistence: a norm scores high when it resists reform and continues to be reproduced across institutional generations. This paper argues that persistence-as-fitness conflates two distinct phenomena that the EPT framework itself requires to distinguish. The first is genuine evolutionary fitness: a norm persists because it continues to win selection competitions in its environment, delivering adaptive value to the replicators that carry it. The second is documentary inertia: a norm persists because the transmission infrastructure, codification, digitization, citation automation, and archival selection, carries it forward independently of its current fitness. Talkie-1930, a 13-billion-parameter language model trained exclusively on English-language text published before January 1, 1931, provides an unusually clear natural experiment for this distinction. The model's documented failure modes, including post-1930 knowledge leakage through editorial metadata, RLHF-induced anachronisms, and corpus contamination by later-dated glosses, demonstrate that no historical corpus constitutes a clean selective environment: every archived text is already the extended phenotype of multiple overlapping transmission processes, not a window onto the original meme pool. The

paper derives three consequences for EPT methodology: a validity constraint on CLI's historical comparative applications, a distinction between persistence by fitness and persistence by transmission inertia as two measurable variables, and a proposal for Inertia-Corrected CLI (IC-CLI) that attempts to disaggregate the two components. A connection is established with the Zombie Law concept developed in prior EPT work: zombie norms may be artifacts of documentary inertia rather than evidence of fitness, which changes both the theoretical interpretation and the policy prescription.

I. INTRODUCTION: THE PERSISTENCE PROBLEM

The Extended Phenotype Theory of law rests on an analogy between biological and cultural evolution that is more than metaphorical. Memes, the units of cultural information that replicate through imitation, transmission, and institutional reproduction, are subject to selection pressures that favor some variants over others. What survives in the legal corpus is, on this account, what fitness favored: norms that delivered adaptive value to the actors who carried and reproduced them. This is the theoretical foundation of the Constitutional Lock-in Index, which treats persistence across institutional generations as a proxy for evolutionary fitness, and of the Zombie Law concept, which identifies norms that persist despite having lost their original adaptive function.

The analogy has a structural vulnerability that the biological version of the theory does not share. In biological evolution, fossil records present their own interpretation challenges, but the mechanisms of preservation are relatively well characterized: mineralization, amber entrapment, and sedimentary layering each introduce known biases. In cultural evolution, and specifically in legal systems, the preservation mechanisms are radically more complex, more opaque, and more active. A legal norm does not simply survive or not survive; it is actively reproduced by a dense infrastructure of codification, citation, commentary, digitization, and archival selection. This infrastructure has its own selection pressures, its own biases, and its own dynamics. A norm that is reproduced by this infrastructure may or may not be fit in the evolutionary sense; it may simply be embedded in a transmission machinery that carries it forward regardless of its current adaptive value.

The distinction between these two sources of persistence, fitness and inertia, has been implicit in EPT work from the beginning but has not been formalized. This paper attempts that formalization, using an unusual empirical anchor: Talkie-1930, a large language model trained exclusively on text published before 1931, whose documented failure modes expose the structure of documentary transmission in unusually clear terms. The model's failures are, from the perspective of this paper, not defects to be corrected but data about how the corpus was constructed and what that construction obscures.

The paper proceeds as follows. Section II describes Talkie-1930 and its relevant failure modes. Section III develops the theoretical distinction between fitness and documentary inertia within the EPT framework. Section IV connects the distinction to niche construction theory as a complementary mechanism. Section V derives the validity constraints that the distinction imposes on CLI. Section VI proposes the Inertia-Corrected CLI as a methodological response. Section VII connects the analysis to the Zombie Law concept. Section VIII draws conclusions and identifies implications for future EPT empirical work.

II. TALKIE-1930: THE MODEL AND ITS INSTRUCTIVE FAILURES

Talkie-1930 is a 13-billion-parameter autoregressive language model developed by Nick Levine, David Duvenaud, and Alec Radford, trained on approximately 260 billion tokens of English-language text published before January 1, 1931: books, newspapers, scientific journals, patents, and legal materials. A substantial portion of the corpus derives from the Harvard Law School Library Institutional Data Initiative, which provided access to nearly one million digitized public-domain volumes. The training corpus cutoff was chosen partly for intellectual property reasons: works published in 1930 entered the United States public domain on January 1, 2026, coinciding with the model release. The cutoff was also a deliberate experimental boundary for the study of generalization versus memorization in temporally constrained language models. The model weights were released under an Apache 2.0 license; compute was provided by Anthropic. An instruction-tuned variant, talkie-1930-13b-it, was also released.

The model's designers acknowledged several failure modes that are directly relevant to the theoretical argument of this paper. The most documented is temporal leakage: despite the pre-1931 training cutoff, the model retains awareness of post-1930 events. A predecessor 7-billion-parameter version demonstrated clear knowledge of Roosevelt's presidency and New Deal legislation. The 13-billion-parameter version retains residual awareness of World War II, the United Nations, and the post-war division of Germany. This leakage did not result from deliberate inclusion of post-1930 material; it resulted from the structure of the pre-1930 corpus itself.

Three mechanisms produced the leakage. First, editorial apparatus in digitized pre-1930 texts: introductions, footnotes, annotations, and metadata added by later editors who were describing the historical document from a post-1930 vantage point. Second, texts that described anticipated futures: pre-1930 publications that discussed projected developments in ways that, once those developments occurred, constitute implicit post-hoc knowledge when read by a model. Third, the n-gram classifier used to exclude post-1930 material was insufficiently granular to catch

embedded anachronisms in multipage documents where early pages were pre-1930 but later pages contained later material.

The second failure mode is RLHF-induced anachronism. The instruction-tuned variant was trained using AI-generated feedback to align the model's outputs with instructed behavior. The 7-billion-parameter version emerged from this process generating responses in listicle format, a rhetorical structure that did not exist in 1930. The post-training process introduced selection pressures from 2025 into a model whose knowledge base was nominally frozen in 1930, producing a hybrid that speaks with 1930's vocabulary and 2025's format.

The third failure mode is corpus bias in the original selection. The pre-1931 corpus was not a representative sample of thought or language in 1920s Anglophone society; it was a sample of published, digitized, and archivally preserved text. Published text in that period was systematically produced by and for the literate class of English-speaking societies, with corresponding over-representation of colonial, eugenicist, and class-stratified perspectives. Users reported that model outputs reflected these biases. The model's designers acknowledged them in the product disclaimer. What the disclaimer does not note is that these biases are not properties of 1920s values in general; they are properties of the transmission infrastructure that selected which 1920s texts would be published, preserved, and digitized.

These three failure modes collectively demonstrate a single structural point: the corpus on which Talkie-1930 was trained is not a window onto the original meme pool of 1920s Anglophone culture. It is the extended phenotype of that meme pool as transformed by a century of editorial selection, archival practice, digitization priorities, and metadata addition. The model trained on this corpus is trained on the transmission infrastructure's output, not on the original environment's input.

III. FITNESS AND DOCUMENTARY INERTIA: A THEORETICAL DISTINCTION

The EPT framework defines memetic fitness as the capacity of a replicator to persist across generations of the selective environment. A legal norm is fit when it continues to be reproduced by the actors and institutions whose behavior it governs, because it continues to deliver adaptive value to those actors and institutions. This is the standard evolutionary account, and it grounds the CLI's use of persistence as a fitness proxy.

Documentary inertia is a structurally different phenomenon. A norm exhibits documentary inertia when it is reproduced not because actors and institutions continue to find it adaptive but because the transmission infrastructure, the set of mechanisms through which norms are codified,

disseminated, cited, and archived, carries it forward independently. The distinction is analogous to the biological distinction between selection and drift: in biological populations, alleles can increase in frequency through random genetic drift rather than fitness advantage. Documentary inertia is the cultural analogue: a meme can persist through transmission drift rather than selection advantage.

III.1 Sources of Documentary Inertia

Four mechanisms generate documentary inertia in legal systems, each distinct from evolutionary fitness. The first is codification lock-in: once a norm is codified in a statute or code, its reproduction is guaranteed by the legislative process itself. Removing a codified norm requires active legislative effort; maintaining it requires no effort at all. This asymmetry means that codified norms persist by default, regardless of whether they continue to deliver adaptive value. The CLI captures this as procedural reform difficulty, but does not distinguish between a norm that resists reform because it is actively valued and one that resists reform because reform is costly and no coalition has formed to demand it.

The second is citation automation. In legal systems where precedent is formally or informally binding, older norms are cited because they are cited in the authorities that cite them. This recursive citation structure can propagate norms that have lost their original adaptive function indefinitely, as long as the citation chain remains unbroken. Automated legal research tools amplify this mechanism: a norm that appears frequently in digitized precedent will be surfaced repeatedly by search algorithms regardless of its current fitness.

The third is digitization selection. The norms and texts that were digitized early and are therefore most accessible in contemporary legal databases are not a representative sample of the historical corpus; they are a sample selected by the priorities, resources, and technical capacities of the institutions that digitized them. Norms embedded in frequently digitized texts have a persistence advantage that is entirely an artifact of digitization infrastructure.

The fourth, most directly illustrated by Talkie-1930, is editorial accumulation. Historical legal texts are embedded in layers of commentary, annotation, and contextual apparatus added by successive generations of editors. A fifteenth-century norm that has been commented on by sixteenth, seventeenth, and eighteenth-century jurists carries with it the documentary weight of those accumulated commentaries. It appears to persist across generations because it does persist, but its persistence is partly a function of the accumulated infrastructure of commentary rather than its continued adaptive value.

III.2 The Conflation Problem for CLI

The CLI measures persistence across institutional generations using five components: procedural reform difficulty, doctrinal entrenchment, organizational lock-in, enforcement capture, and TCI-linked transmission capacity. Each of these components is correlated with documentary inertia as well as with evolutionary fitness. A norm that scores high on doctrinal entrenchment

may be entrenched because it is fitness-superior in its environment (genuine lock-in) or because it is embedded in a citation network that reproduces it independently of fitness (inertia). The CLI cannot distinguish these cases because it does not contain an inertia-correcting component.

The conflation is not merely a measurement problem; it is a theoretical problem. If documentary inertia and evolutionary fitness produce the same CLI scores, then CLI-based conclusions about why norms persist are underdetermined. The claim that Argentina's labor law persists because it has achieved lock-in fitness in the existing selective environment is compatible with the alternative claim that it persists because it is embedded in a citation and codification infrastructure that reproduces it regardless of fitness. The two accounts have different policy implications, different predictions about when the norm will change, and different connections to the broader EPT theoretical framework.

IV. NICHE CONSTRUCTION AND TRANSMISSION INFRASTRUCTURE

The distinction between fitness and documentary inertia connects to niche construction theory, which the EPT programme has identified as a complementary framework for understanding how institutions modify their own selective environments (Lerer 2024, *Law as Constructed Niche*, forthcoming). In biological niche construction theory, developed by Odling-Smee, Laland, and Feldman (2003), organisms do not merely adapt to pre-existing environments; they modify those environments in ways that alter the selection pressures their descendants will face. The beaver's dam is the paradigm case: it transforms the local hydrological environment, changing the fitness landscape for beavers and for many other organisms in that environment.

Applied to legal norms, niche construction suggests that successful memes do not merely survive their selective environment; they modify it. A norm that achieves codification has modified its environment by creating a legal infrastructure that makes its own removal costly. A norm that generates a body of supportive precedent has modified its environment by creating citation networks that amplify its presence in subsequent legal reasoning. A norm that becomes the subject of academic commentary has modified its environment by creating an interpretive community whose expertise is invested in the norm's continuation.

The niche construction perspective reframes the fitness-inertia distinction. Documentary inertia is not an alternative to fitness; it is, in part, the extended phenotype of past fitness. A norm that was genuinely fit in an earlier environment used that fitness to construct a transmission infrastructure that now carries it forward even when the original fitness advantage has dissipated. The residual persistence of a formerly fit norm through its constructed niche is not the same as the persistence of a currently fit norm in a competitive environment, but it is not entirely independent

of fitness either. The infrastructure was built by fitness; its continued operation is what inertia describes.

Talkie-1930 illustrates this niche construction dynamic with unusual clarity. The texts that appear in the pre-1930 corpus are not a random sample of 1920s discourse; they are the texts that were fit enough to be published, significant enough to be preserved, and accessible enough to be digitized. Each of these selection stages was itself a competitive process in which only some texts survived. The surviving texts then constructed a niche: the digitized corpus that became the training environment for Talkie-1930. The model's behavior is the extended phenotype of that constructed niche, not of the original 1920s meme pool. The editorial apparatus that produced the leakage failures is precisely the accumulated niche-construction work of a century of editors, archivists, and digitizers.

For EPT methodology, this suggests that the transmission infrastructure is not merely a neutral medium through which memes flow; it is itself an extended phenotype with its own dynamics. Measuring a norm's persistence without characterizing the transmission infrastructure through which it persists conflates the norm's current fitness with the persistence of the infrastructure it previously constructed. The IC-CLI proposed in Section VI attempts to address this conflation directly.

V. VALIDITY CONSTRAINTS ON CLI'S HISTORICAL APPLICATIONS

The CLI was validated against a dataset of 60 cases with $AUC = 0.97$, demonstrating strong predictive validity for contemporary cross-national comparisons (Lerer 2024). The theoretical argument of this paper does not challenge this validation; it imposes a scope constraint on what the CLI can legitimately claim to measure in specific application contexts.

Three validity constraints follow from the fitness-inertia distinction.

V.1 The Historical Comparison Constraint

When CLI is used to compare the persistence of a norm across historical periods within the same system, for example, to track changes in Argentina's labor law CLI score between 1990 and 2026, the comparison assumes that the transmission infrastructure has remained constant across the comparison period. If the infrastructure changed, for example through digitization of legal databases in the 1990s, expansion of online legal research tools in the 2000s, or automation of citation tracking in the 2010s, then changes in CLI scores may reflect changes in documentary inertia rather than changes in evolutionary fitness. A norm that appears to have become more

entrenched may simply have become more thoroughly embedded in the digital citation infrastructure.

This constraint does not invalidate historical CLI comparisons; it requires that they be accompanied by an analysis of how the transmission infrastructure changed across the comparison period. Where infrastructure change was substantial, the comparison must be qualified accordingly.

V.2 The Cross-System Comparison Constraint

When CLI is used to compare the persistence of norms across different legal systems, for example, to compare Argentina's labor law CLI score with Brazil's, the comparison assumes that the transmission infrastructures of the two systems are comparable in their inertia-generating properties. Systems with more extensive digitization, more automated citation, and more formally binding precedent have higher inertia-generating infrastructure, independently of the fitness of the norms they contain. A norm in a high-infrastructure system may achieve a higher CLI score than an equally fit norm in a low-infrastructure system, simply because the infrastructure is more efficient at generating persistence.

This constraint is particularly relevant for comparisons between common law systems, which have extensive formally binding precedent, and civil law systems, which rely more heavily on codification. The two systems have structurally different inertia-generating properties, and direct CLI comparisons between them require infrastructure-correcting adjustments.

V.3 The Zombie Law Interpretation Constraint

The Zombie Law concept identifies norms that persist despite having lost their original adaptive function. The concept presupposes that persistence in the absence of fitness is detectable; if all persistence is partly inertial, then Zombie Law is a limiting case of a continuous variable rather than a binary category. The constraint this imposes is interpretive: a high CLI score combined with evidence of lost adaptive function is consistent with two distinct diagnoses. The norm may be a genuine Zombie: formerly fit, now inert but preserved by lock-in mechanisms. Or it may be a documentation artifact: never particularly fit, but embedded in a high-inertia transmission infrastructure that made it appear to have been selected when it was only carried. The two diagnoses have different implications for reform strategy. A genuine Zombie requires disrupting the lock-in mechanisms. A documentation artifact requires only correcting the transmission infrastructure's selection biases.

VI. TOWARD AN INERTIA-CORRECTED CLI

The three validity constraints do not require abandoning CLI as a measurement instrument. They require supplementing it with a component that proxies for documentary inertia independently of evolutionary fitness. The Inertia-Corrected CLI (IC-CLI) proposed here is a first approximation; its operationalization requires empirical development that this paper does not attempt to complete.

The IC-CLI decomposes the standard CLI score into two components: a fitness-attributable component and an inertia-attributable component. The fitness component captures persistence that is associated with active reproduction by actors and institutions that find the norm adaptive. The inertia component captures persistence that is attributable to the transmission infrastructure independently of active reproduction.

VI.1 Candidate Inertia Indicators

Four candidate indicators proxy for documentary inertia independently of fitness. The first is citation automation density: the proportion of citations to a norm in recent legal materials that appear in automatically generated bibliographies, case management systems, or citation-tracking outputs, rather than in deliberately authored arguments. High citation automation density suggests that a norm is being carried by the infrastructure rather than actively chosen by legal actors.

The second is commentary accumulation ratio: the ratio of secondary commentary (academic articles, treatises, law review notes) referencing the norm to primary legal materials (statutes, regulations, court decisions) applying it. A norm with a high commentary accumulation ratio has more documentary infrastructure than active application; this pattern is consistent with inertial persistence.

The third is digitization age advantage: the number of years by which the norm's primary text was digitized earlier than the average for texts of comparable age in the same legal system. Early digitization creates citation network advantages that compound over time; controlling for this advantage separates transmission-infrastructure effects from fitness effects.

The fourth is reform attempt frequency: the number of documented legislative or regulatory proposals to amend or repeal the norm in a given period, regardless of outcome. A norm with high reform attempt frequency is being actively contested, which suggests that its persistence reflects successful resistance to selection pressure (fitness) rather than absence of selection pressure (inertia). A norm with low reform attempt frequency may simply not have attracted the attention needed to generate reform proposals, which is consistent with inertial persistence beneath the threshold of political salience.

VI.2 The IC-CLI Formula

The IC-CLI is defined as:

$$\text{IC-CLI} = \text{CLI} - w(\text{I})$$

where I is a composite inertia index constructed from the four candidate indicators above, and w is an empirically estimated weight that converts the inertia index into CLI-equivalent units. The subtraction yields the fitness-attributable component of persistence after removing the inertia-attributable component.

This formulation is deliberately simple. The empirical challenge is estimating w , which requires a dataset in which persistence can be independently attributed to fitness or inertia through means other than the CLI itself. One candidate source is natural experiments created by legal system transitions: when a legal system undergoes wholesale replacement, as occurred in Eastern European countries after 1989, the norms that survive the transition and are actively reproduced by new actors provide a fitness-attributable persistence measure that can be compared against the norms that persist only in archived texts but are not applied. The ratio of these two persistence measures in the post-transition period provides an empirical basis for calibrating w .

A second candidate source is the Talkie-1930 failure mode data itself. The documented leakage events, in which post-1930 knowledge entered the model through editorial apparatus rather than through training data, provide a measurable proxy for documentary inertia in the pre-1930 corpus: the proportion of the model's post-1930 knowledge that is attributable to editorial apparatus rather than to genuine pre-1930 content. This proportion is, in principle, estimable through ablation studies that remove editorial apparatus from the training data and measure the resulting change in temporal leakage. The ratio provides a direct empirical measure of the inertia-to-fitness ratio in that specific corpus.

VII. DOCUMENTARY INERTIA AND THE ZOMBIE LAW RECONSIDERED

The Zombie Law concept, as developed in prior EPT work, identifies norms that persist in the formal legal corpus despite having lost their original adaptive function. They continue to appear in statutes, to be cited in pleadings, and to be reproduced in legal databases, but they are not applied in ways that reflect genuine normative force. They are, in the evolutionary analogy, vestigial structures: present in the phenotype, no longer contributing to fitness.

The fitness-inertia distinction proposed in this paper requires a refinement of the Zombie Law diagnosis. The original concept assumes that Zombie persistence is the result of lock-in mechanisms that prevent the selection environment from eliminating a formerly fit norm. The documentary inertia analysis adds a second pathway: a norm may appear zombie-like not because it survived selection despite losing fitness, but because it was never primarily selected at all. It persists because it was embedded early in a high-inertia transmission infrastructure, and that infrastructure has carried it forward through documentary momentum.

This distinction matters practically. The standard EPT prescription for Zombie Law is disruption of the lock-in mechanisms that prevent selection from operating: making reform procedurally easier, weakening the coalition that benefits from the norm's persistence, introducing competitive pressure from alternative norms. This prescription assumes that if the lock-in is removed, selection will eliminate the zombie. But if the persistence is primarily inertial rather than lock-in based, removing formal lock-in mechanisms may not be sufficient: the norm will continue to be reproduced by the citation and digitization infrastructure even if it is no longer formally binding. Correcting an inertially persistent norm requires interventions in the transmission infrastructure itself: active removal from legal databases, explicit deprecation in citation guides, and curriculum changes in legal education.

Talkie-1930 makes this concrete. A model trained to remove post-1930 knowledge by simply filtering training data will continue to exhibit leakage because the pre-1930 texts themselves contain post-1930 information in their editorial apparatus. The intervention must reach the infrastructure, not just the content. The parallel for legal reform is direct: a legislature that repeals a zombie norm but does not address its embeddedness in the citation infrastructure will find the norm continuing to be reproduced in legal arguments, contracts, and regulatory decisions for years after formal repeal.

VIII. CONCLUSION: THE CORPUS IS NOT THE ENVIRONMENT

The central claim of this paper can be stated simply: in cultural evolution, the archive is not the environment. The corpus that survives to the present day is the extended phenotype of the transmission infrastructure, not a window onto the original selective environment. Training a model on the surviving corpus, whether it is a language model learning 1920s English or an EPT researcher measuring 1990s CLI scores, means training on the output of a century of editorial selection, digitization priorities, and citation network dynamics. The fitness signals embedded in that corpus are real, but they are confounded with inertia signals that have nothing to do with current or original fitness.

Talkie-1930 makes this visible because the temporal boundary is sharp enough to create measurable leakage. The leakage is not a technical failure; it is a demonstration that even a well-resourced team with a clean theoretical boundary cannot produce a corpus that accurately represents a historical meme pool. Every text carries the sediment of its transmission history. Every model trained on that text learns the sediment along with the signal.

For the EPT programme, the implications are methodological before they are theoretical. The CLI is a valid instrument for what it measures: the composite persistence of a norm in its

current institutional environment, including both its fitness-attributable and inertia-attributable components. The validity constraints proposed in this paper do not undermine the CLI; they specify the conditions under which its scores can be interpreted as fitness measures rather than composite persistence measures. Meeting those conditions requires the IC-CLI extension and the empirical development of inertia indicators that this paper has only sketched.

The deeper implication is theoretical and connects to the EPT programme's core commitment to evolutionary mechanism. If persistence is the proxy for fitness, and if persistence conflates fitness with inertia, then the programme's explanatory claims are systematically ambiguous between two accounts that have different mechanisms, different predictions, and different policy implications. Resolving that ambiguity is not a refinement of the programme; it is a precondition for the programme's scientific credibility. Talkie-1930 did not create this problem. It made it impossible to ignore.

LAKATOSIAN EPISTEMOLOGICAL DECLARATION

This paper operates within the Extended Phenotype Theory of law, constituted as a Lakatosian progressive research programme (Lakatos 1978). The hard core: legal institutions are extended phenotypes of cultural replicators subject to evolutionary selection; the relevant unit of analysis is the replicator and its fitness landscape; institutional change follows memetic selection dynamics. The protective belt includes the specific methodological claims advanced here (fitness-inertia distinction, IC-CLI, inertia indicators), which may be modified without abandoning the hard core. The paper functions as a negative heuristic intervention: it identifies a measurement problem within the programme's existing empirical instruments and proposes a correction. The correction is itself falsifiable: if the IC-CLI's inertia component does not predict persistence patterns differently from the standard CLI component in systems with documented infrastructure differences, the inertia-correction hypothesis is refuted. Mandatory citations: Lakatos (1978); Popper (1959); Hull (1988); Campbell (1974).

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AI DISCLOSURE

Artificial intelligence tools were used for literature synthesis, source identification, and drafting assistance. The theoretical framework, the fitness-inertia distinction, the IC-CLI proposal, all validity constraints, the Zombie Law refinement, and all original arguments are the author's own contribution. The author accepts full responsibility for all content and errors.